SECTION 7 · SUBSECTION 7.1
STARTING AND OPERATING

7.1.1 INTRODUCTION.

There are two deployment shapes and they differ in more than transport. The local shape runs the API in one process and executes jobs on a thread pool inside it. The distributed shape adds Redis and at least one worker process; the API only enqueues. Both share the same store interface and the same pipeline function. For reading or testing the codebase, the local shape needs no Redis and no model key if the LLM stage is skipped (figure 7.1-1).

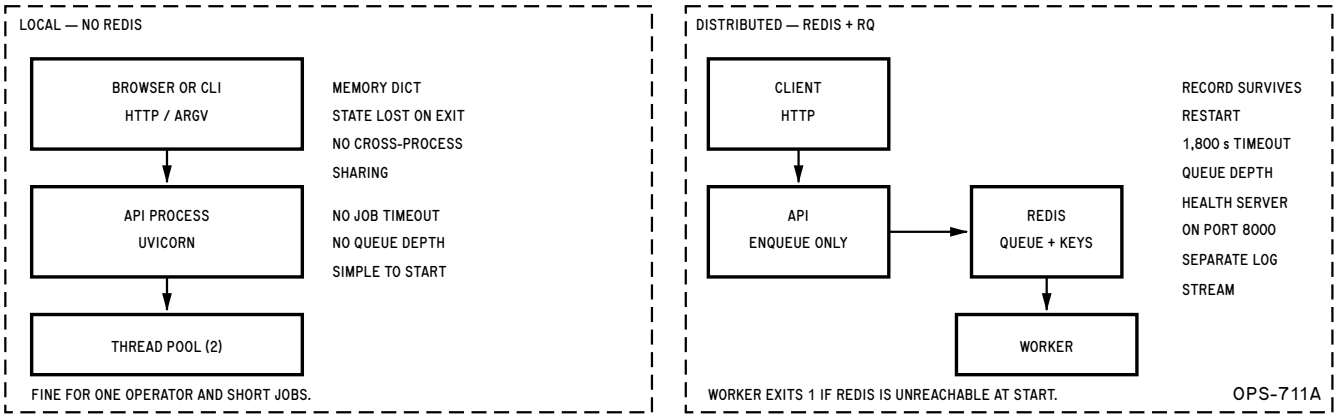


Figure 7.1-1. Process Topology With and Without Redis

7.1.2 FUNCTIONAL DESCRIPTION.

A CLI run needs an interpreter, dependencies, a GitHub address and, unless *--skip-llm* is passed, an NVIDIA key. It writes ten files into the output directory. An API run needs the same environment plus an HTTP client; when Redis is absent, the API executes the pipeline in a two-thread local pool. The frontend is independent: it needs Node, a build, and the API origin.

Environment is loaded from *.env* by *python-dotenv* in both entry points. Variable names below come from the example file only; no values were read for this handbook. A private repository requires a token, which is propagated per request and never persisted in a job record.

7.1.2.1 Commands Executed for This Handbook.

The commands below were run in the working tree on the change date. Nothing else was executed: in particular, no remote analysis, no Docker build, no worker start and no frontend build. Results are reported exactly as observed.

Table 7.1-1. Verified Commands and Observed Results

Command	Observed result
.venv\Scripts\python.exe -m pytest tests -q	476 passed in 20.40 s
.venv\Scripts\python.exe -m pytest --collect-only -q	476 cases across 15 files
.venv\Scripts\python.exe run.py --help	Usage printed; exits 0; options *--url*, *--token*, *--incremental*, *--output*, *--no-html*, *--skip-llm*, *--max-explain*, *--test-rag*
.venv\Scripts\python.exe -c "import src.api.main"	Imports without error; no startup side effects observed
python -m uvicorn src.api.main:app --port 8011	Startup complete; listener on 127.0.0.1:8011
GET http://127.0.0.1:8011/health	*status ok*, version 1.0.0, active_jobs 0, redis_connected false, queue_mode local_threadpool, depth 0
Interpreter and browser resolution	Python 3.11.15 in *.venv*; Chrome present at its standard Windows path

7.1.3 PERFORMANCE AND DESIGN DATA.

Table 7.1-2. Environment Variable Names

Name	Required	Effect
NVIDIA_API_KEY	for LLM/RAG	Embeddings and explanations
NVIDIA_BASE_URL	no	Endpoint override
NVIDIA_MODEL	no	Chat model override
GITHUB_TOKEN	no	Private repos; 5,000 req/h
CHROMA_DB_PATH	no	Persistence directory
REDIS_URL	no	Enables Redis mode
RQ_QUEUE_NAME	no	Default *gnosis_jobs*
JOB_TIMEOUT_SECONDS	no	Default 1,800
JOB_TTL_SECONDS	no	Result expiry
MAX_*_ANALYSIS_FILES	no	Tier thresholds
PORT	worker	Health listener
VITE_API_URL	frontend	API origin at build time

CAUTION

Not executed: *docker compose up*, *python -m src.api.worker*, *npm ci*, *npm run build*, and any analysis of a real repository. Their documented behaviour is described earlier from code, but this handbook makes no claim that they succeed on this machine.

```
START CHECK                                  OPERATOR
1 RUN - pytest, expect 476
2 RUN - run.py --help
  √ START - /health returns ok
3 READ - queue_mode and redis flag
4 STOP - terminate listener
```

OPS